

Sowmiya S

Machine Learning Enthusiast | Data Science & Full-Stack Developer

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EDUCATION

B.Tech – Information Science & Engineering

Women's Engineering College (Govt), Puducherry

Relevant Coursework: Machine Learning, Data Structures & Algorithms, Database Management, Computer Networks

Sep 2023 – Jun 2027

CGPA: 7.8 / 10

SKILLS

- **Languages:** Python, Java, C, SQL, R
- **ML / DL:** TensorFlow, PyTorch, Scikit-learn, Pandas, NumPy, Matplotlib
- **Web:** HTML5, CSS3, JavaScript (ES6+), REST APIs
- **Tools:** MySQL, Git, GitHub, Jupyter Notebook, Google Colab, VS Code, MS Excel
- **Soft Skills:** People Management, Team Collaboration, Communication

WORK EXPERIENCE

Research Intern

Jun 2025 – Jul 2025

National Institute of Technology, Tiruchirappalli (NIT Trichy)

- Researched hyperspectral image classification using CNN and Vision Transformer (ViT) architectures on Indian Pines dataset.
- Achieved 98% classification accuracy, outperforming CNN baseline using spectral-spatial patch extraction with ViT.
- Optimised preprocessing (PCA dimensionality reduction, normalisation) and hyperparameter tuning to elevate training throughput.
- Proposed model achieved 98.73% training accuracy and 98.11% validation accuracy on complex multi-class image classification.

PROJECTS

Hyper Spectral Image Classification – CNN & Vision Transformer

Jun – Jul 2025

- Deep learning pipeline using CNN and ViT on Indian Pines dataset (200+ bands, 10,249 samples). Reached 98% accuracy. Tech: Python, PyTorch, Scikit-learn.

Real-Time Weather Web Application

Feb 2026

- Responsive single-page weather portal using OpenWeatherMap API with async/await, live forecasts, and dynamic DOM updates. Tech: HTML, CSS, JavaScript.

Frontend Web Development Portfolio

May 2025 – Mar 2026

- Designed and developed multi-page interactive web portals including company profile, event registration, and e-commerce showcase. Tech: HTML5, CSS3, JavaScript.

CERTIFICATIONS

- **Diploma in Full Stack Web Development** (Self-paced / Offline)
- **Biological Data Analysis & Visualisation with R** (NPTEL – IIT Kharagpur)
- **Java Programming for Beginners** (Online Course)
- **Design, Technology & Innovation** (NPTEL – IIT Bombay)
- **Research Contribution Certificate** (NIT Tiruchirappalli)
- **Diploma in Computer-aided Teaching** (Offline)

Nov 2023 – Nov 2024
Feb 2025 – Mar 2025
Mar 2025 – Apr 2025
Jun 2025 – Sep 2025
Jun 2025 – Jul 2025
Apr 2023 – Oct 2023

LEADERSHIP & ACTIVITIES

Student Organiser – Lunova '26 Technical Symposium

Feb 2026

- Coordinated logistics and execution for Women's Engineering College 1st inter-departmental symposium, managing 20+ volunteers across 5 departments.

Class Representative – 3rd Year (2025–2026)

2025 – 2026

- Primary liaison between 40+ students and faculty, resolving academic scheduling and organizing student events.